



## Technology Overview

Engineered to safeguard IoT and IIoT devices, the Terafence **TFG-1URP** employs Air-Gap technology to segment and isolate industrial end-devices from malicious or unauthorized malware attacks, ensuring complete protection against threats.

At the core of the solution is Terafence's proprietary FPGA-based hardware architecture, developed and manufactured by **Terafence** which establishes a fully controlled data path between two network segments. While allowing normal protocol data to flow unidirectionally, the return path is entirely eliminated—effectively creating an Air-Gapped environment.

Acting as a Protocol Proxy, Terafence terminates TCP/IP sessions at both ends while transferring only raw data between the two unidirectional gateway interfaces. Since raw data is never stored within the unit, there is no need for encryption or additional data protection measures. As a result, Terafence eliminates the reliance on cryptographic security mechanisms and instead enforces security by preventing network access altogether.

Beyond protecting IoT and IIoT end-devices from cyber threats, the **TFG-1URP** also safeguards other network assets by preventing compromised devices from spreading malware, executing cyberattacks, or causing operational disruptions.

### Supported Protocols

**TFG-1URP** supports multiple, simultaneous industrial and IT protocols, including:

- **SFTP/FTP/SMB** – Secure file or folder-tree transfers
- **SMTP** – Email relay
- **SYSLOG Forwarding** – Log event transmission
- **RTSP/ONVIF** – Live streaming for CCTV
- **HTTP/S** – File and data uploads to web servers or cloud platforms
- **Local SYSLOG Forwarding** – Internal log event management
- **Modbus** – Communication between multiple PLCs and HMIs
- **IEC-104** – Protocol for power utilities
- **MQTT** – Data brokering and publishing
- And various other protocols as per customer requirements.

## Why choose TFG-1URP?

### Key Features:

- Complete galvanic **network isolation** for enhanced security and protection.
- Proprietary **Terafence FPGA-based hardware** architecture for **hardware-enforced security**.
- Two dedicated CPUs for **protocol support, processing, and termination**.
- **Secure HTTPS-based web GUI** for configuration and management, accessible only from the secure side.
- **Hardened** operating system on the dedicated CPUs.
- Core security hardware operates independently **without** an operating system, MAC address, or IP address.

### Technical Specifications:

- **Data Throughput:** Up to 1 Gbps with auto-negotiation
- **Power Supply:** 100–240 VAC, 50/60 Hz, with dual redundant internal hot-swappable AC power supplies
- **Network Interfaces:** 2 × Ethernet (RJ-45) ports
- **Form Factor:** 1U rack-mountable appliance
- **Mounting Options:** 19-inch 1U rack mount
- **Operating Temperature:** 32°F to 113°F
- **Storage Temperature:** -40 °F to 185 °F
- **Design:** Fanless architecture with no moving mechanical parts for enhanced reliability
- **Compliance:** FCC, CE, and EMI/EMC

### Solution Highlights:

- Delivers **physical network isolation** through Terafence's proprietary **FPGA-based** hardware architecture.
- Supports **protocol processing** and termination through two dedicated proxy engines.
- Provides secure and intuitive configuration through a restricted **HTTPS-based** management interface.
- Designed for high availability, long-term operation, and a **high MTBF**.
- Certified to **IEC 62443** and **MIL-STD-810F** standards.